Year 12 Further Maths - Roots of polynomials

Similar questions - New equations by substitution and symmetric functions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The cubic equation $x^3 - 2x^2 + 5x - 3 = 0$ has roots α, β and γ. By making an appropriate substitution, or otherwise, find a cubic equation with integer coefficients whose roots are $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$. **[3]**

2 The equation $3x^3 - x^2 + 4x + 2 = 0$ has roots α, β and γ. Use a substitution to find a cubic equation with integer coefficients whose roots are $\alpha + 2$, $\beta + 2$ and $\gamma + 2$. **[4]**

3 In this question you must show detailed reasoning. The roots of $4x^3 - 3x^2 + x + 6 = 0$ are α, β and γ. Find a cubic equation with integer coefficients whose roots are $\alpha\beta$, $\beta\gamma$ and $\gamma\alpha$. **[6]**

4 The roots of the equation $x^4 - 3x^3 + 2x - 5 = 0$ are α, β, γ and δ. By using a suitable substitution, find a quartic equation whose roots are $\alpha - 1$, $\beta - 1$, $\gamma - 1$ and $\delta - 1$, giving your answer in the form $ay^4 + by^3 + cy^2 + dy + e = 0$ with integer coefficients. **[5]**

Question 1
Let $y = \frac{1}{x}$, $x = \frac{1}{y}$: $\frac{1}{y^3} - \frac{2}{y^2} + \frac{5}{y} - 3 = 0$. Multiplying by y^3: $1 - 2y + 5y^2 - 3y^3 = 0$, i.e. $3y^3 - 5y^2 + 2y - 1 = 0$

Question 2
Let $y = x + 2$, $x = y - 2$: $3(y-2)^3 - (y-2)^2 + 4(y-2) + 2 = 0$. Expanding: $3y^3 - 18y^2 + 36y - 24 - y^2 + 4y - 4 + 4y - 8 + 2 = 0$, i.e. $3y^3 - 19y^2 + 44y - 34 = 0$

Question 3
$\sum\alpha = \frac{3}{4}$, $\sum\alpha\beta = \frac{1}{4}$, $\alpha\beta\gamma = -\frac{3}{2}$. New roots: sum $= \sum\alpha\beta = \frac{1}{4}$. Sum of products in pairs $= \alpha\beta\gamma(\alpha+\beta+\gamma) = -\frac{3}{2} \times \frac{3}{4} = -\frac{9}{8}$. Product $= (\alpha\beta\gamma)^2 = \frac{9}{4}$. Equation: $y^3 - \frac{1}{4}y^2 - \frac{9}{8}y - \frac{9}{4} = 0$, i.e. $8y^3 - 2y^2 - 9y - 18 = 0$

Question 4
Let $y = x - 1$, $x = y + 1$: $(y+1)^4 - 3(y+1)^3 + 2(y+1) - 5 = 0$. $y^4 + 4y^3 + 6y^2 + 4y + 1 - 3y^3 - 9y^2 - 9y - 3 + 2y + 2 - 5 = 0$, so $y^4 + y^3 - 3y^2 - 3y - 5 = 0$